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Integration of Viscosity/Density Measurements into an Untethered Logging Tool for Rapid Downhole Polymer Degradation Monitoring of EOR Fluids

Miguel Gonzalez, Huseyin Seren, Tim Thiel, Hichem Abdelmoula, Robert Adams, and Arun Mahadev, Aramco Services Company: Aramco Research Center–Houston, Houston, Texas, USA; Subhash Ayirala and Mohammed Alotaibi, Exploration and Petroleum Engineering Advanced Research Centre, Saudi Aramco, Dhahran, Saudi Arabia; Rami Jabari and Max Deffenbaugh, Aramco Services Company: Aramco Research Center–Houston, Houston, Texas, USA

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Abstract

Enhanced Oil Recovery (EOR) relies on polymer fluids that are injected to improve oil mobility and reservoir sweep efficiency, enabling improved recovery rates. Polymer degradation, however, can pose serious challenges to the operation if not properly addressed during the injection process. Here we propose using an untethered logging device with an on-board viscometer for direct downhole assessment of polymer degradation and show preliminary tests of the integrated system in a water well. This tool would enable cost-effective and on-demand measurements not currently available to operators. The tool consists of a compact, untethered, and buoyancy-controlled logging tool hosting an onboard viscosity/density sensor. The tool is a handheld device featuring a magnetically attached weight used to make the tool negatively buoyant. The weight pulls the tool down to the bottom of the well. When a target depth is reached, the magnetically attached weight is released, allowing the tool to float back up to the surface. During the descent/ascent period, density and viscosity are logged by use of a custom quartz tuning fork resonator, acquiring continuous measurements. Pressure and temperature data are also obtained with other onboard sensors. Tests of the tool were done in a 1450 ft water well. A series of tests spanning across three different depths were performed. In one of the tests, the logging tool was programmed to remain downhole for about 20 minutes. Logs of viscosity and density were obtained by a calibrated quartz tuning fork sensor at a rate of one sample every ten seconds. The sensor was calibrated using Newtonian fluids before the test. In prior studies, the sensor was also tested using Xanthan Gum and other polymer solutions at different concentrations to calibrate the sensor for use in non-Newtonian fluids [1,2]. Full logs of the viscosity/density of the fluids along the water well were obtained. While the density and viscosity held relatively constant, as expected, since the well also hosts other contaminant fluids from prior operations, changes in fluid properties were seen to vary. This was verified by out-of-well tests of the system in a pressurized vessel, where only minimal changes in fluid properties were observed. The viscosity/density sensor was then demonstrated to be quite

sensitive, making it promising to detect small changes in fluid properties if properly packaged to avoid contamination.

Introduction

Enhanced oil recovery (EOR) operations make use of polymers to viscosify injection water in order to better facilitate both the oil displacement as well as the sweep efficiency in the reservoir, both improving oil recovery. During injection, however, these polymers can undergo mechanical deformation or chemical changes that induce degradation [3, 4, 5]. When a polymer fluid degrades, the molecular weight of the macromolecules within the fluid is reduced, thereby reducing the fluids viscosity. This reduction in viscosity, can therefore compromise the effectiveness of the EOR process [6].

Currently, the injected fluids are designed in the laboratory and prepared in the field accordingly. Their properties, however, may be different than those in the laboratory, or may change during the fluid injection. Laboratory rheometers exist to verify that the obtained viscosity in the field is as designed, however, there are no tools that can help field operators assess the quality of the fluid directly downhole. Different logging tools have incorporated viscometers for formation evaluation applications [7, 8, 9]. However, standard logging operations required large crews and complicated operations to deploy a sensor. If a viscosity measurement is required periodically, this could incur in large costs and overly complicated logistics.

Here, we demonstrate the use of an untethered logging platform [10, 11] with an onboard miniature quartz tuning fork (QTF) resonator as the viscometer [12, 13]. In this work, we show results on the integration of the two and preliminary tests in a water well. In future tests, the combined system will be used for logging of fluid viscosity at injection wells without the need of a wireline logging operation, enabling more frequent and on-demand measurements of viscosity for the purpose of assessing whether polymer degradation has occurred downhole during EOR operations.

Tools Description

The tuning fork device implemented in the current tool and some preliminary measurements have been described elsewhere [1]. Here, we show an upgraded tool designed for well testing and explain how the sensor and tool components were packaged.

To ensure reliable conveyance within the well, the tuning fork system is mounted on the untethered tool. This setup guarantees the consistent operation of the tuning fork. Importantly, all electrical components are encapsulated, providing insulation from well fluids and ensuring their protection.

The overall dimensions of the system used for the tuning fork are shown in Figure 1.

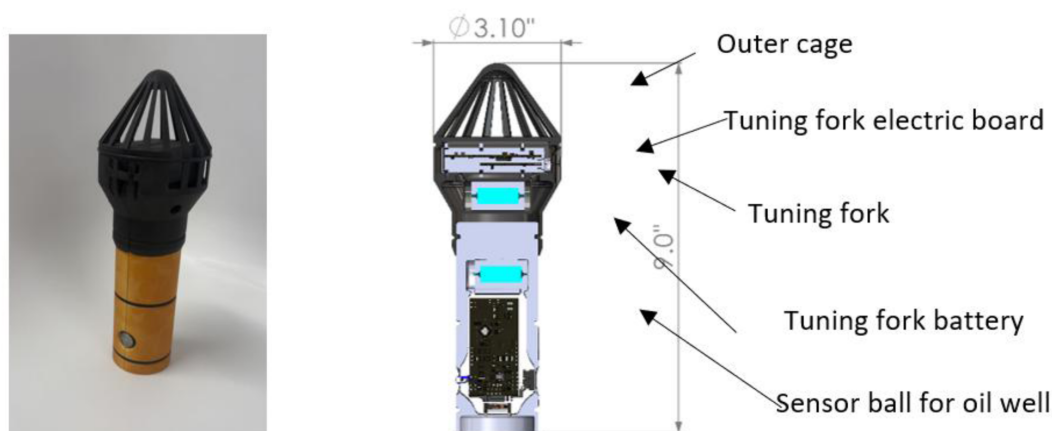


Figure 1—Dimensions of the tuning fork on untethered tool used for well test

The tuning fork system consists of the following main components that protect and ensure safe deployment of the measurement.

- **The tuning fork:** the system's sensor, is encapsulated separately on a 3D printed part to prevent epoxy from affecting its resonance frequency and Q-value.
- **Electrical boards:** The electrical boards are encapsulated in urethane to protect against well fluid. They record the tuning fork's impedance vs frequency behavior and save it in flash memory.
- **Encapsulated battery:** The encapsulated battery powers tuning fork boards. The encapsulation protects the battery from well fluids and external pressure.
- **Sensor outer cage:** This cage secures the encapsulated electronics and battery on the untethered tool. It consists of two parts joined by O-rings and protects the tuning fork from impact damage against well walls.

Figure 2 shows images of the above-described components.

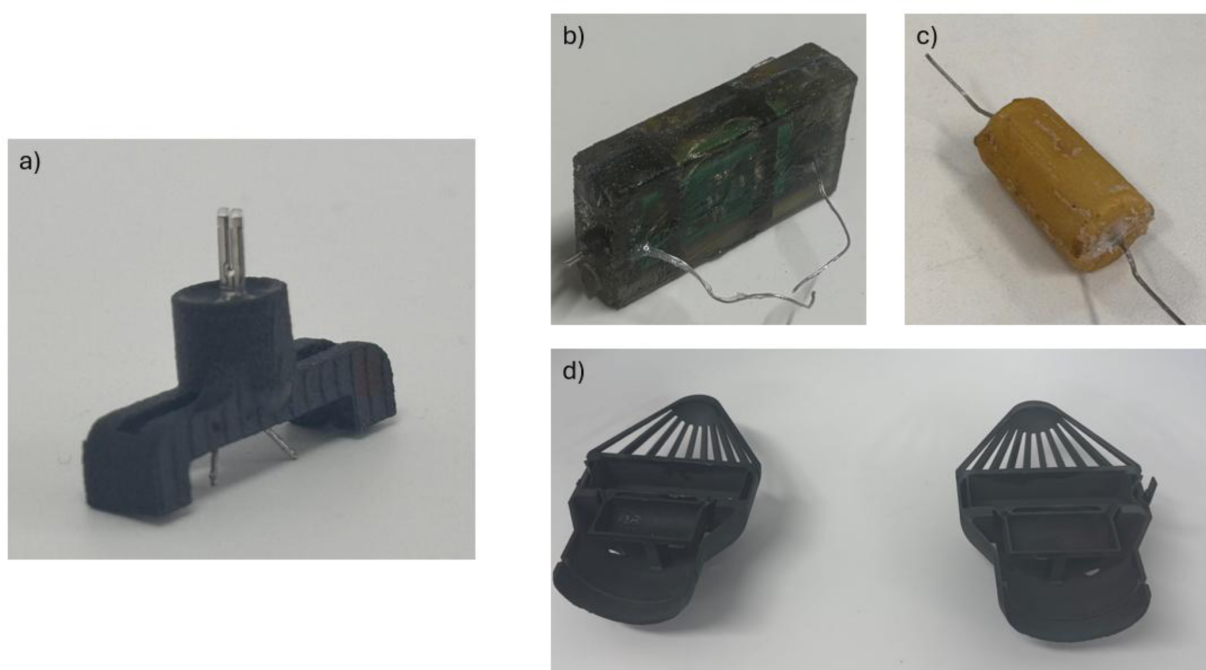


Figure 2— a) Tuning fork sensor encapsulated in 3D printed carrier. b) Encapsulated electrical board with tuning fork circuitry. c) Encapsulated battery to withstand high pressure and temperature. d) Outer cages to house encapsulated electronics and battery.

Well Test Results

A series of tests of the tool with onboard tuning fork viscometer were performed in a water well. The well has a total depth of 1450 ft. A total of three tests were performed. In the first test the tool went down to ~820 ft, while on the second test the tool went down to ~1300ft. In the third test, the tool was programmed to sit at the bottomhole. The tool was timed to release the weight in 1 hour so that after it reached total depth it would reside at the bottom of the well for approximately 20 minutes collecting data, after which the weight was released.

Well Test 1

The tool was initially programmed to release its magnetically attached weight 14 minutes after mission time or when reaching 580 psi. Figure 3 shows images of the deployed and recovered system. The tool was released into the well close to 13 minutes before the expected weight release. The estimated descent

time was ~ 1 ft/sec, while the actual descent time was ~ 0.4 ft/s. Figure 4 shows plots for temperature and pressure versus time. Figure 5 shows logs for pressure/temperature/magnetic-field sensors versus depth. The total depth was estimated from the hydrostatic pressure using the pressure sensor and validated using the magnetometer data which was used as a casing collar locator. The total mission time was closer to 45 minutes, indicating a slower ascent or descent time than estimated. While descending/ascending, the tools onboard sensors logged temperature, pressure, and magnetic field. The tuning fork system logged viscosity and density using the tuning fork sensor.



Figure 3—(Left) Tool with onboard viscometer during deployment. (Right) Recovered system after well test.

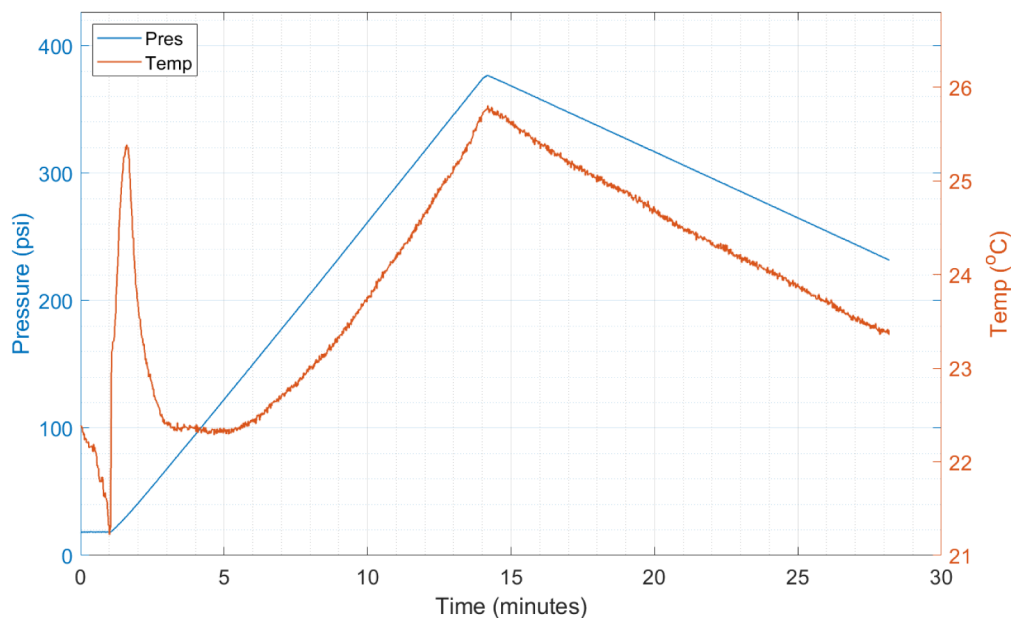


Figure 4—Pressure/temperature versus time curve for well test 1.

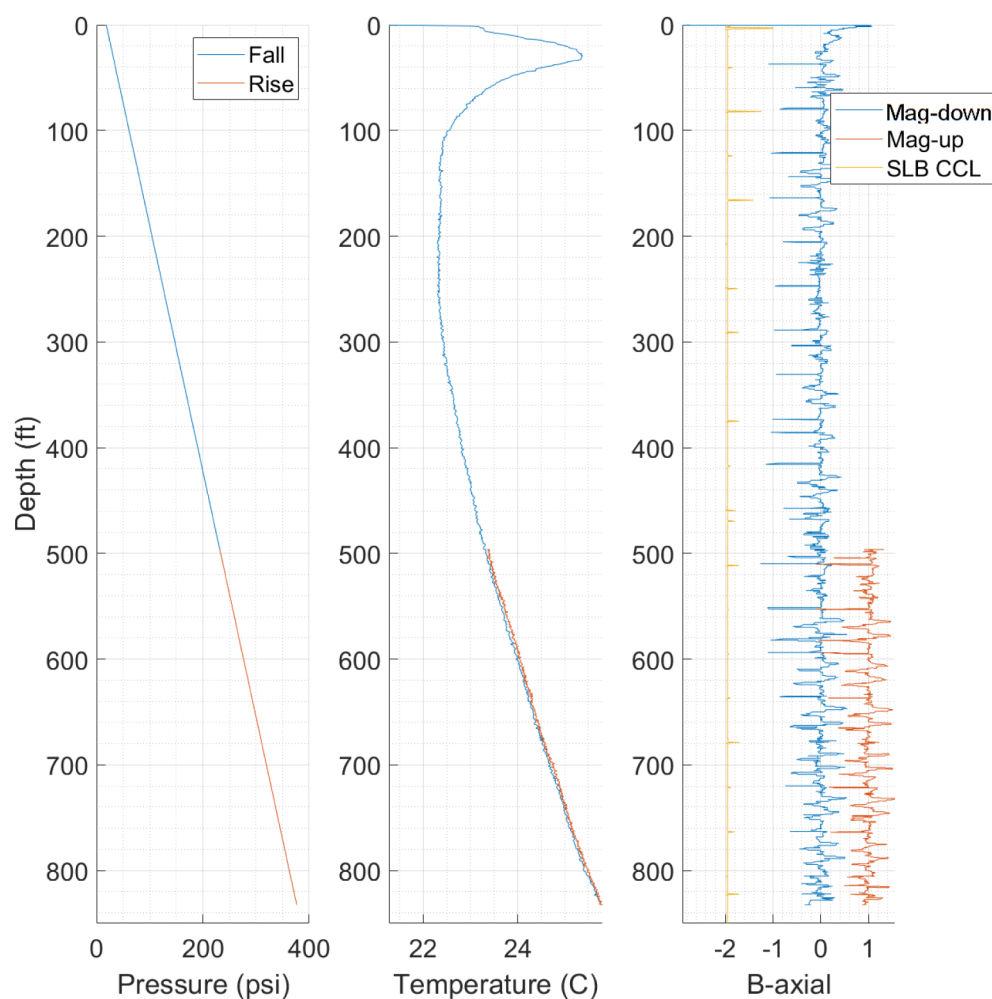


Figure 5—(Left) Log curve for pressure versus depth. (Center) Log curve for temperature versus depth. (Right) Log curve for magnetic field versus depth.

Figure 6 shows a plot of the obtained data from the calibrated onboard tuning fork. In this deployment, because the tuning fork system was decoupled from the tool's sensors, we cannot correlate the depth obtained with the tool with the obtained viscosity/density data. However, fork measurements were taken every 1 minute for this test and the moment of entry and retrieval can in principle be obtained from the jumps in the data at the beginning and end of the viscosity/density curves, as seen in Figure 6. The fork was calibrated in two Newtonian fluids: water and isopropyl alcohol. The calibration coefficients used for this fork were $C_p = 2.73$ and $C_\mu = 0.0485$. For f_{air} we use a value of 31486.79 Hz. The data within the shaded blue region represents the data obtained within the well. The density and viscosity correspond to that of water, as expected. The data shown in the green region shows data when the tuning fork is in air right before and after the deployment in the water well. An interesting trend in increasing viscosity was observed right immersion in the water well and continued close to the half point of the test. After the second part of the test, likely corresponding with the ascent of the tool, the viscosity seems to have plateaued.

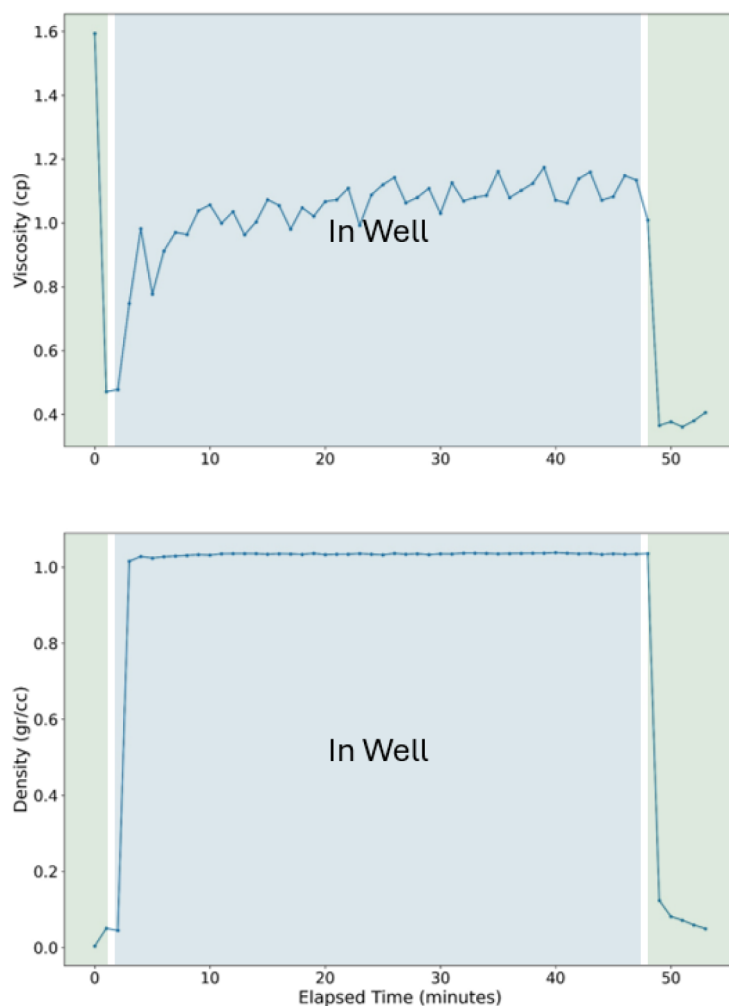


Figure 6—Viscosity and density obtained during well test. The shaded blue region contains the data points obtained when the tool was in the well. The shaded green area are measurements in air before and after recovery of the tool.

Well Test 2

For this test the tool was program to release the weight when reaching a pressure of 580 psi plus a 20 psi offset to prevent the log from stopping at the surface. A timer for a 1hr release time was also programed. The tool was released into the well 6 minutes after the timer started. Figure 7 shows plots for the pressure, temperature, and magnetic fields taken with the untethered tool as a function of time, while Figure 8 shows the logs versus depth.

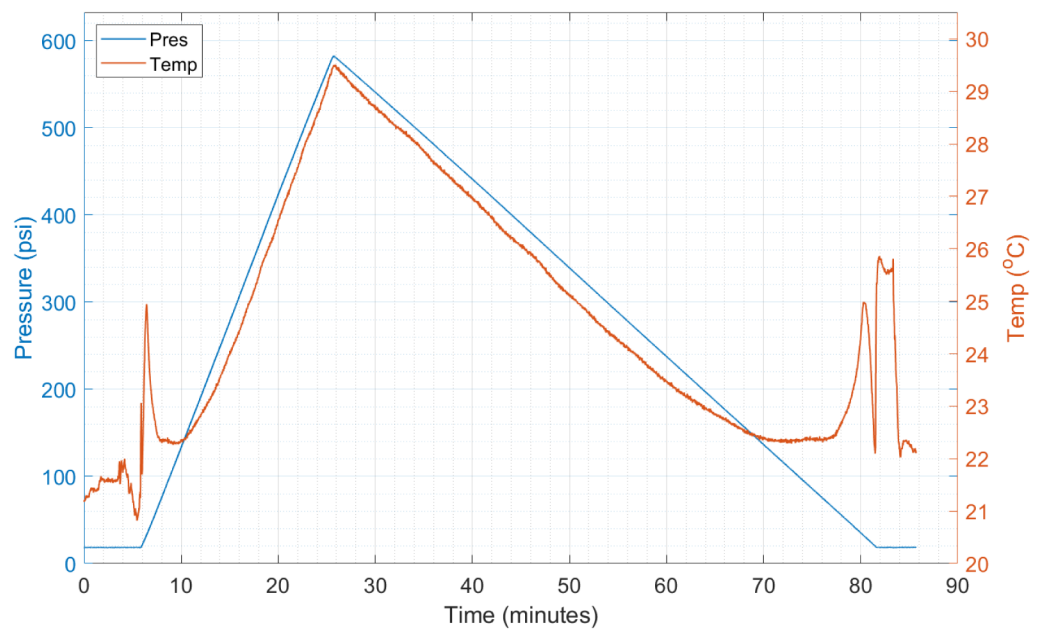


Figure 7—Pressure/temperature versus time curve for well test 2

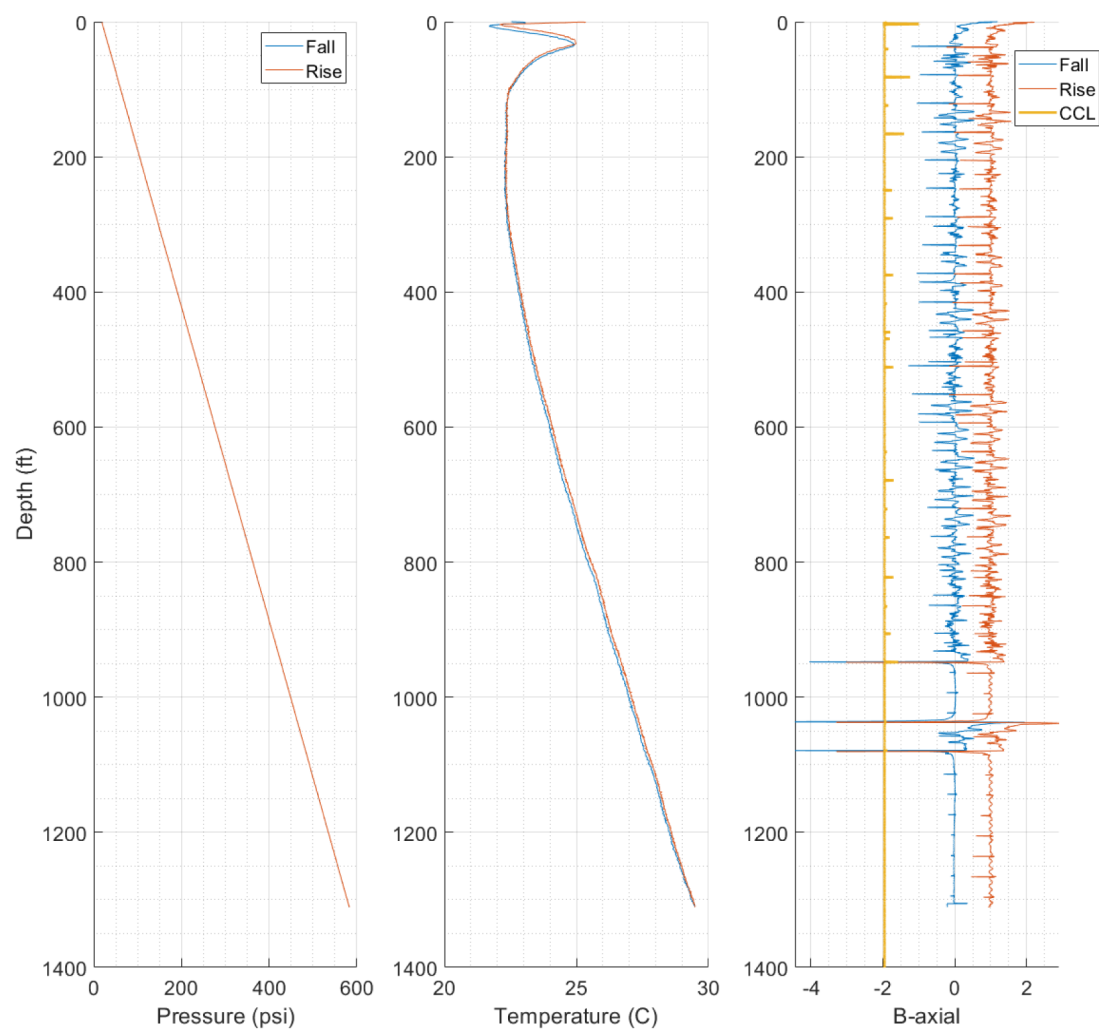


Figure 8—(Left) Log curve for pressure versus depth. (Center) Log curve for temperature versus depth. (Right) Log curve for magnetic field versus depth.

A total depth of 1300 ft was reached on this test. Figure 9 shows the data for the viscosity and density obtained with the tuning fork. In this test, the measurement frequency was increased significantly to one measurement every 10 seconds. This was done in order to decrease our measurement noise by averaging more data points. Figure 10 shows the smoothed data using a rolling average with a size 10 window.

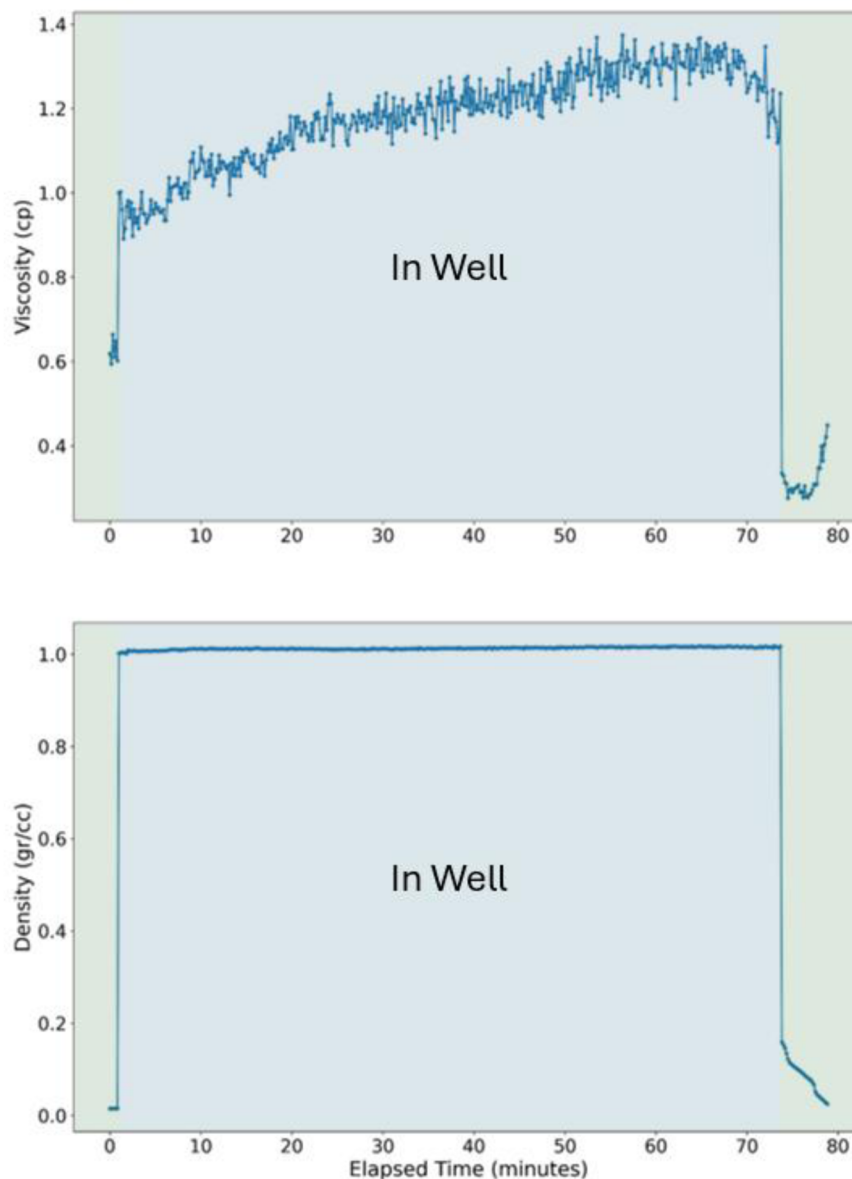


Figure 9—Viscosity and density obtained during well test 2. The shaded blue region contains the data points obtained when the tool was in the well. The shaded green area are measurements in air before and after recovery of the tool.

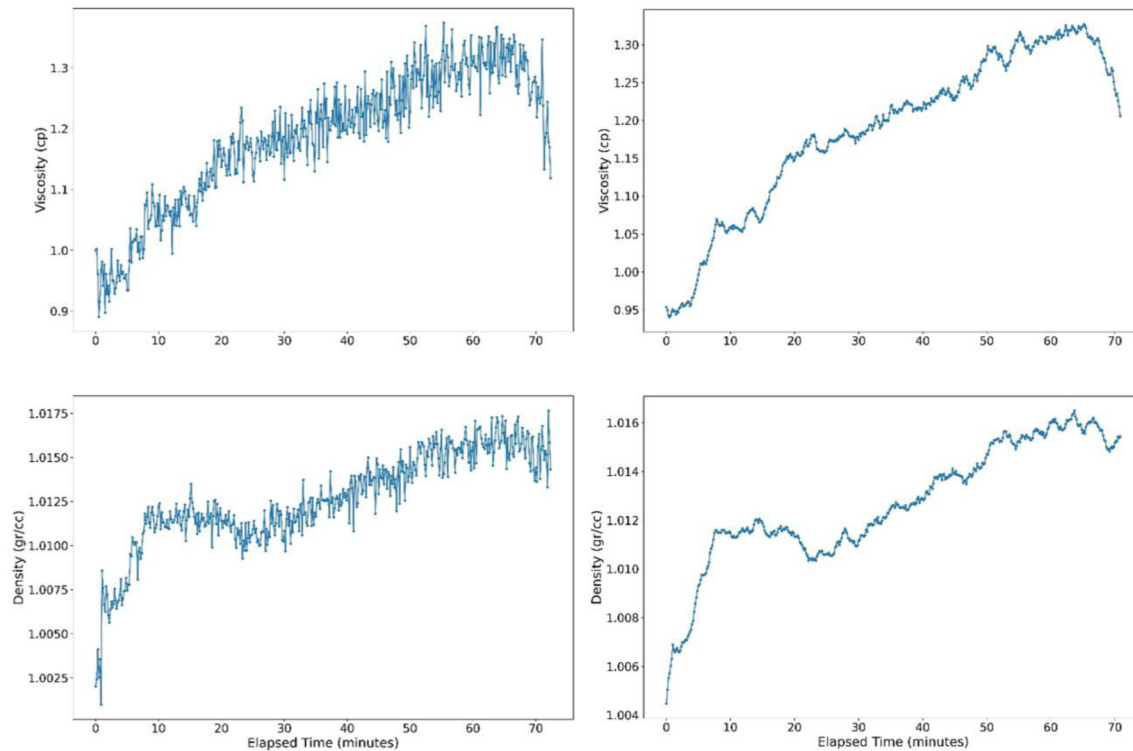


Figure 10—Viscosity and density obtained during well test # 2. On the left side, viscosity (top) and density (bottom) while in the well without averaging. On the right side, viscosity (top) and density (bottom) while in the well and smoothed using a rolling average with an averaging window of 10 measurements.

The calibration coefficients used for this fork were $C_p = 2.73$ and $C_\mu = 0.1$. For f_{air} we use a value of 31486.79 Hz. We noticed that for this run, the viscosity calibration needed to be adjusted to a larger value. It is possible that the prior calibration was done with a fork that was not properly cleaned, compensating with a lower calibration coefficient the increased peak width from the dirty fork. After well test 1, the fork was cleaned with isopropyl alcohol. The coefficient $C_\mu = 0.1$ is more in line for a typical calibration coefficient as seen in most previous devices. In Figure 9, as before, the data within the shaded blue region represents the data obtained within the well. The density and viscosity correspond to that of water, as expected, but with an increasing drift as previously observed. The data shown in the green region shows data when the tuning fork is in air right before and after the deployment in the water well.

Well Test 3

In this last test a semi-permanent type of deployment was implemented for the tool. For this case, weight release trigger was based on time. The time to release the weight was selected such that the tool can sit at the bottom of the well for a prolonged time.

Figure 11 shows a pressure/temperature versus time curve for the entire mission. It can be seen that approximately 20 minutes after tool deployment, the tool reached total depth and stayed there for another 20 minutes. After that, the weight was released and the tool returned to surface in approximately 1 hour. The logs of the tool versus depth are shown in Figure 12.

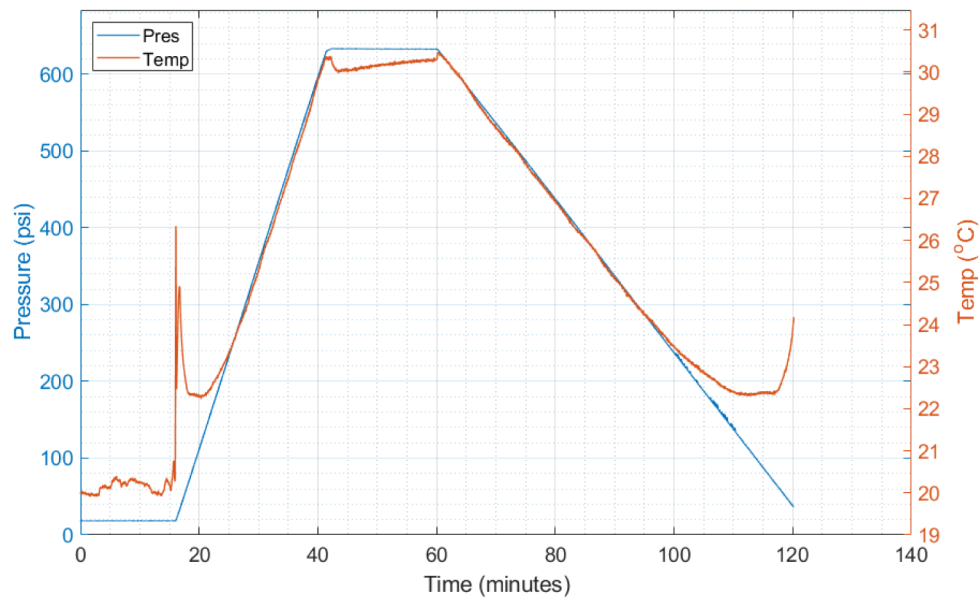


Figure 11—Pressure/temperature versus time curve for semi-permanent tool deployment for continuous viscosity/density measurements at the bottom of the well.

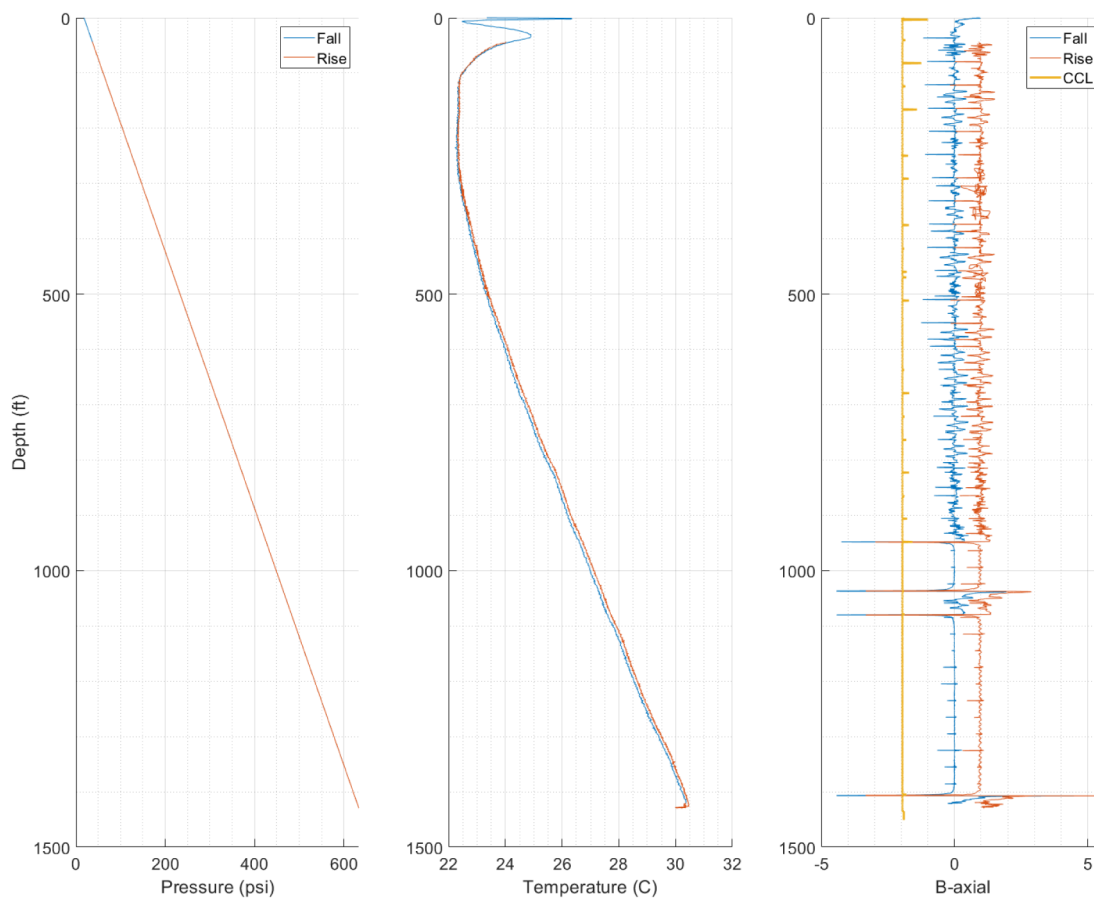


Figure 12—(Left) Log curve for pressure versus depth. (Center) Log curve for temperature versus depth. (Right) Log curve for magnetic field versus depth.

The viscosity/density measurements are shown in Figure 13, which includes the raw data and the smooth data obtained using a rolling average as in well test 2. The calibration coefficients used for this fork were $C_p = 2.73$ and $C_\mu = 0.1$. For f_{air} we use a value of 31486.79 Hz. These are the same coefficients used for well test

#2. In this test, again we see an increasing trend in viscosity as a function of mission time. The reason for this trend is not completely understood at the moment but it is believed that, because the effect is cumulative in time, it might be due to erratic mass accumulation from well debris which introduces extra damping on the fork. Images of the fork taken after well test 3 show significant accumulation of debris on the fork.

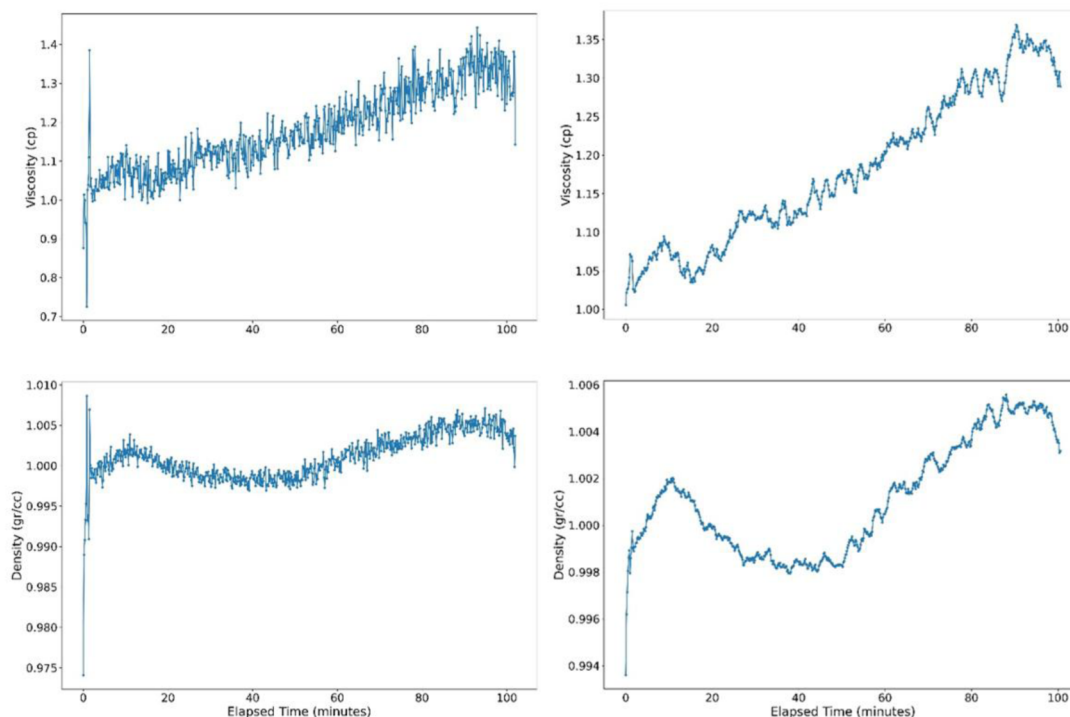


Figure 13—Viscosity and density obtained during well test # 3. On the left side, viscosity (top) and density (bottom) while in the well without averaging. On the right side, viscosity (top) and density (bottom) while in the well and smoothed using a rolling average with an averaging window of 10 measurements.

Pressure Chamber Test

This test was carried out using a hydrostatic test chamber filled with hydraulic oil to transfer pressure to the tool/tuning fork system. The tool was placed inside a sealed bag containing water so that the test fluid is the same as the well test. The pressure was slowly ramped up and down, from atmospheric up to 650 psi, similar to the bottom of the water well. The first ramp up in pressure was done in approximately 13 minutes, while the ramp down was done in 7 minutes. A couple of extra quick ramps up and down were done afterwards with a combined up/down ramp of less than 5 minutes each.

Figure 14 shows the plots for the viscosity and density measurements obtained during the pressure chamber test. On the left of the figure, the un-averaged raw data is shown. No increasing trend is seen here from the raw viscosity data. The density data shows trends that match the pressure in the chamber, i.e. the density increases when the pressure increases and decreases when the pressure decreases. The peaks for the three pressure ramps can be clearly seen centered at close to 10 minutes, 19 and 22 minutes. These peaks can be more clearly seen in the averaged data on the right of Figure 14, where again a rolling average with window of 10 measurements was used. For the averaged data, the viscosity also shows some small variations that match the pressure changes in the chamber, however, these changes are only a few percent, while the changes in the well tests were up to 30%. This leads us to conclude that pressure has a minimal effect on the measurement and that it is more likely that the drift in viscosity observed during the well test is due to the accumulation of debris from the well on the fork. New sensor packaging strategies would have to be devised in order to address this issue for future deployments.

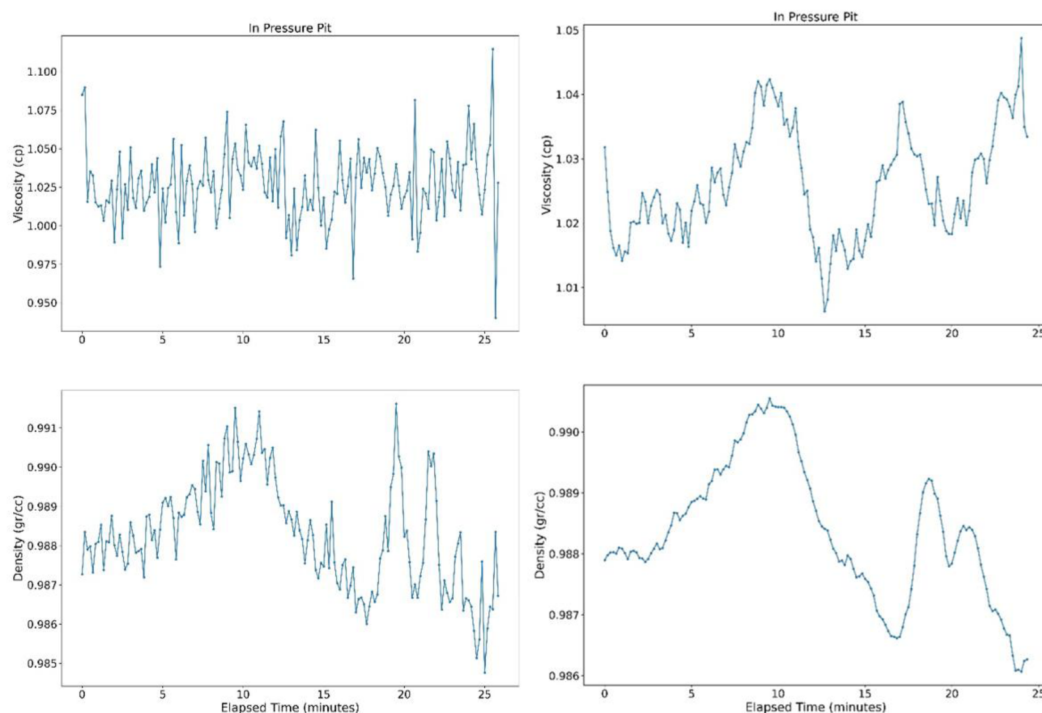


Figure 14—Viscosity and density obtained during pressure chamber test. On the left side, viscosity (top) and density (bottom) while in the well without averaging. On the right side, viscosity (top) and density (bottom) while in the well and smoothed using a rolling average with an averaging window of 10 measurements.

Conclusions

We have presented a novel untethered logging tool in this study for downhole viscosity measurements. The tool, consisting of a buoyancy-controlled enclosure and a customized quartz tuning fork resonator was demonstrated in the lab environment for obtaining logs of viscosity from liquid wells. The tool was then successfully deployed for measurements of viscosity and density in a water well, demonstrating the full integrated tool capabilities. Three deployments of the tool were conducted at different well depths to obtain the full logs of the viscosity/density. A hydrostatic pressure chamber test has also been carried out by simulating the same fluid characteristics (as in the well tests) to understand the effects of pressure on fluid density and viscosity. The major findings from these tests are summarized below in the following.

1. Well test#1: Relatively accurate viscosity data may be obtained with low measurement frequency data (measurements every minute). This can be advantageous for long missions or if trends in viscosity changes are the target and the tool's power and memory consumption needs to be conserved.
2. Well test#2: If higher frequency measurements are needed, the measurement frequency can be increased, in this case by performing one measurement every 10 seconds. This would be useful if high resolution viscosity might be needed or if averaging is necessary to detect more subtle trends in the data. A hybrid approach could be implemented where data is logged at a low frequency and then the system switches to a more accurate high frequency acquisition at a depth of interest.
3. Well test#3: Here, we demonstrated a semi-permanent deployment system for the untethered tool that sits at the bottom of the well. This enabled us to monitor viscosity/density at bottomhole while maintaining the tool with a safe distance from any debris as the fork was placed at the top of the tool. In this case, viscosity and density were monitored for about 20 minutes.
4. Pressure chamber test: Here we verified that the observed variations in viscosity and to a lesser extent in density, were not due to pressure effects. In this case the observed changes are hypothesized to be due to debris accumulation on the fork as the tool travels down the well and back.

There are currently no tools that can be used to detect degradation downhole in an efficient and low-cost manner and the presented tool could potentially be used for frequent downhole monitoring of polymer degradation during EOR operations thanks to its compact size and easy deployment characteristics.

Future improvements on the system will include integrating the tuning fork measurement into the untethered tool control board, improved packaging of the sensor to avoid sensor fouling, and integration of additional sensors such as conductivity and other electrochemical sensing devices for a more complete multiparameter measurement.

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